From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE (Main), 2014

CONCEPT RECAPITULATION TEST - IV

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

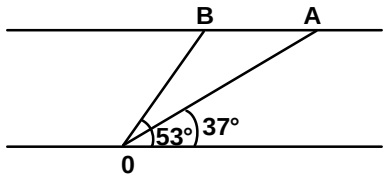
SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. The minimum and maximum magnitude which is possible by adding four forces of magnitudes 1N, 3N, 9N and 10N is
   (A) 0 and 23 N (B) 1N and 23 N
   (C) 2N and 23 N (D) 3N and 23 N

2. A boat crosses a 2.4km wide river to reach the a point A on the opposite bank, located at 37° with the flow, in 500s. The same boat takes the same time (500s) to reach another point B on the opposite bank located at 53° with the flow. Where should a landing point be chosen on the other bank so that the boat can cross in the minimum possible time?



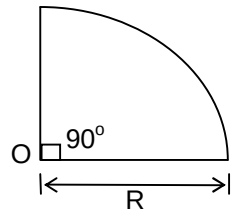
   (A) 1.2 km downstream from O (B) 2.4 km downstream from O
   (C) 3.6 km downstream from O (D) 7.2 km downstream from O

3. A particle is moving in x–y plane. At certain instant, the components of its velocity and acceleration are as follows. V_x = 3m/s, V_y = 4m/s, a_x = 2 m/s^2 and a_y = 1 m/s^2. The rate of change of speed at this moment is
   (A) $\sqrt{10}\,m/s^2$ (B) $4\,m/s^2$
   (C) 10 m/s^2 (D) 2 m/s^2

4. A body of mass 'm' is moving in a circle of radius with a constant speed 'v'. Let work done by the resultant force in moving the body over half circumference and full circumference is W and W' then
   (A) $W = \frac{mv^2}{r}(\pi r),\ W' = \frac{mv^2}{r}(2\pi r)$ (B) $W = \frac{mv^2}{r}(2r),\ W' = \text{zero}$
   (C) $W = \frac{mv^2}{r}(\pi r),\ W' = \text{zero}$ (D) W = 0, W' = 0

Space for Rough work

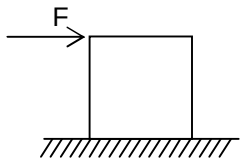
5. In the figure, one fourth part of uniform disc of radius 'R' is shown. The distance of the centre of mass of this object from centre 'O' is



(A) $\frac{4R}{3\pi}$ (B) $\frac{2R}{3\pi}$

(C) $\sqrt{2}\frac{4R}{3\pi}$ (D) $\sqrt{2}\frac{2R}{3\pi}$

6. A force F is applied on the top of a cube as shown in the figure. The coefficient of friction between the cube and ground is μ. If F is gradually increased the cube will topple before sliding if

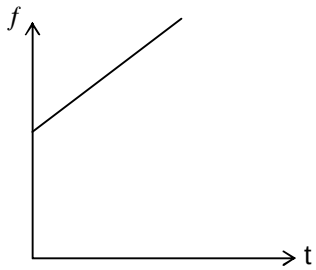


(A) $\mu > 1$ (B) $\mu < \frac{1}{2}$

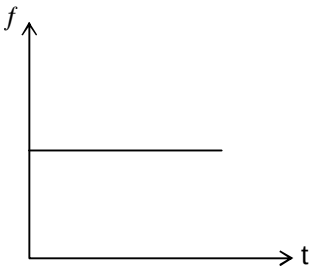
(C) $\mu > \frac{1}{2}$ (D) $\mu < 1$

7. An observer starts moving with uniform acceleration 'a' towards a stationary sound source of frequency f_o. As the observer approaches the source, the apparent frequency 'f' heard by the observer varies with time 't' as

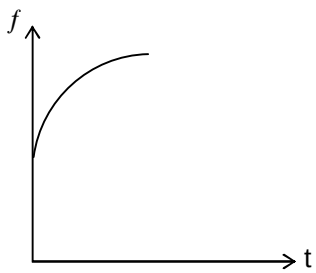
(A)



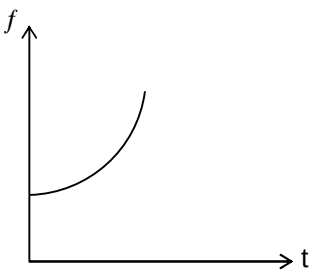
(B)



(C)

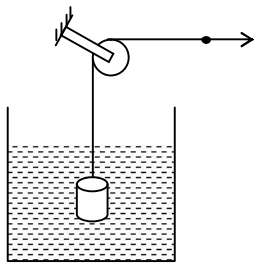


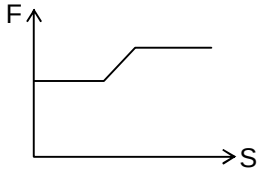
(D)

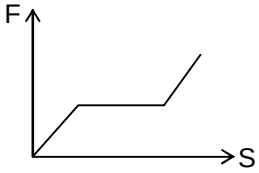


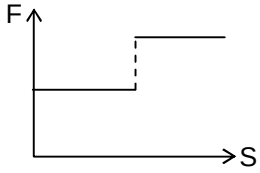
Space for Rough work

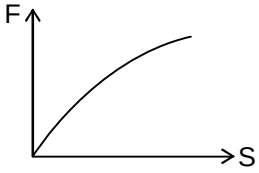
8. A metallic cylinder of radius r and height h is totally submerged in a liquid contained in a beaker as shown in the figure. If force F is pulling the string at a constant velocity v, the graph showing the variation of force F against displacement of cylinder is best represented by



(A) 

(B) 

(C) 

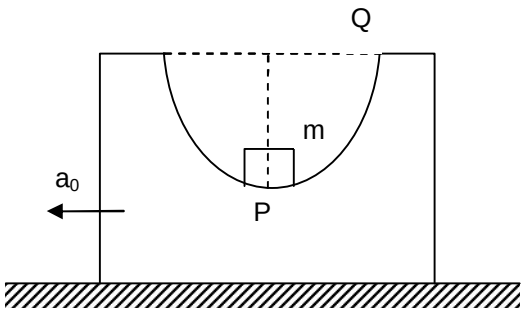
(D) 

9. A particle starts moving with velocity $\vec{v} = \hat{i}$ (m/s), and acceleration $\vec{a} = \left(\hat{i} + \hat{j}\right)$ (m/s^2). Find the radius of curvature of the particle at t = 1sec.

(A) 5 m
(B) $5\sqrt{5}$ m
(C) $6\sqrt{5}$ m
(D) none

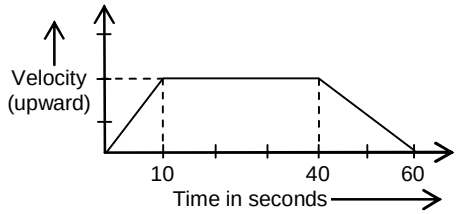
10. A block of mass m is lying at rest at point P of a wedge having a smooth semi-circular track of radius R. What should be the minimum value of a_0 so that the mass can just reach point Q.

(A) $\frac{g}{2}$
(C) g
(D) not possible



Space for Rough work

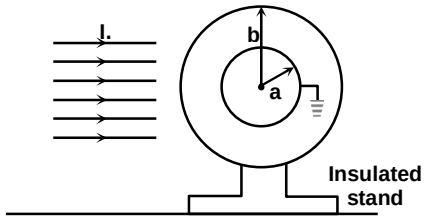
11. A person is standing on a weighing machine placed on the floor of an elevator. The motion of the elevator is shown in the adjacent diagram. The maximum and the minimum weights recorded are 66 kg and 57 kg. The true weight of the person is [Take g = 10 m/s^2.]



(A) 60 kg
(B) 63 kg
(C) 61.5 kg
(D) 65 kg

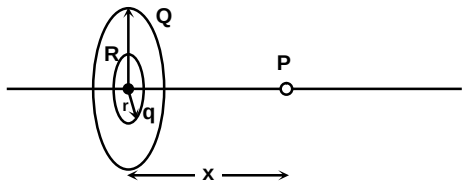
12. An x-ray tube is operating at 150 kV and 10 mA. If only 1% of the electric power supplied is converted into X-ray, the rate at which heat is produced in the target is
(A) 3.55 cal/sec
(B) 35.5 cal/sec
(C) 355 cal/sec
(D) 3550 cal/sec

13. A parallel beam of light of monochromatic radiation of cross-section area A ($<\pi b^2$), intensity I and frequency ν is incident on a solid, perfectly absorbing conducting sphere of work function ϕ_0 ($h\nu > \phi_0$) and radius b. The inner sphere of radius a is grounded by a conducting wire. Assume that for each incident photon one photoelectron is ejected. Current through the conducting wire just after the radiation is



(A) $\frac{IAe}{h\nu}\frac{b}{a}$
(B) $\frac{IAe}{2h\nu}$
(C) $\frac{2IAe}{h\nu}$
(D) $\frac{IAe}{h\nu}\frac{a}{b}$

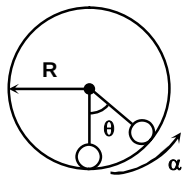
14. Two concentric rings of radius R = 8 cm and r = 2 cm having charge Q = 8 μC and q = -1 μC are placed as shown in diagram. Then electric field is zero at axis of rings



(A) x = $\pm 4\sqrt{3}$ cm
(B) x = $\pm$ 4 cm
(C) x = $\pm$ 2 cm
(D) x = $\pm 2\sqrt{3}$ cm

Space for Rough work

15. A small sphere of mass m and radius r is placed inside a hollow cylinder of radius R. Now cylinder rotating with some angular acceleration which slowly increases from zero to certain value α. There is no slipping between two surfaces during motion.



(A) Maximum angle formed by line joining centre will be $\theta = \sin^{-1}\left(\frac{2\alpha R}{5g}\right)$

(B) Angular velocity of sphere will be $\omega' = \frac{\omega r}{R}$ where ω is angular velocity of cylinder.

(C) sphere will keep rotating at lowest position.

(D) K.E. of sphere becomes constant when it reaches maximum height.

16. Suppose the potential energy between the electron and proton at a distance r is given by $-\frac{Ke^2}{3r^3}$.

Application of Bohr's theory of hydrogen atom in this case shows that energy in the nth orbit is proportional to

(A) n^6 (B) n^{-3}

(C) n^{-2} (D) n^{-4}

17. Block of mass m is placed on a wedge of mass M at height H. All surfaces are smooth and system is released, then which of the following is incorrect.



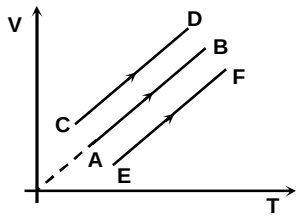
(A) Work done by normal reaction on the wedge is positive and on block is negative.

(B) work done by normal reaction is zero for both the block and the wedge.

(C) change in gravitational potential energy is shared by the block and wedge in inverse ratio of their masses.

(D) Centre of mass will not change position.

18. Following are some processes shown in V-T graph for ideal gas. Which of following is incorrect.



(A) in process CD pressure continuously increases.

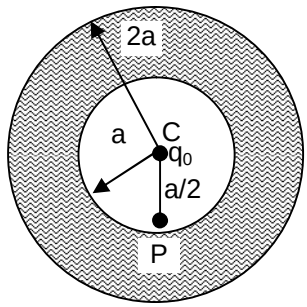
(B) in process EF pressure continuously decreases.

(C) Pressure remains constant in all process AB, CD, EF.

(D) Internal energy of gas increases in all three processes.

Space for Rough work

19. A point positive charge q_0 is placed at a point C inside the cavity in spherical conductor as shown in figure. Find the potential at P.



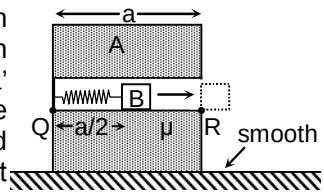
(A) $\frac{Kq_0}{2a}$ (B) $\frac{Kq_0}{a}$

(C) $\frac{2kq_0}{a}$ (D) $\frac{3Kq_0}{2a}$

$\left[\text{here,} k = \frac{1}{4\pi \in_0}\right]$

20. In screw gauge; 5 complete rotations of circular scale give 1.5 mm reading on linear scale. Circular scale has 50 divisions. Least count of the screw gauge is
(A) 0.006 mm (B) 0.003 mm
(C) 0.015 mm (D) 0.03 mm

21. A cubical block 'A' of mass m_1 and edge length 'a' lies on a smooth horizontal floor. It has a groove with a flat base with an open end. On the closed end of the groove there is a spring of natural length 'a' attached to it. A small block 'B' of mass m_2 is pushed into the groove compressing the spring by a/2. Coefficient of friction between B and groove is μ. System is now released from rest. Find speed of B as it comes out of A. Assume that B is always sliding against the surface of the groove. (m_1 = 3kg, m_2 = 1kg, a = 10 cm, k = 100N/m, μ = 0.13)



(A) 15 cm/s (B) 10 cm/s
(C) 5 cm/s (D) 20 cm/s

22. The end of capillary tube is immersed into a liquid. Liquid slowly rises in the tube up to a height. The capillary-fluid system
(A) will absorb heat (B) will release heat
(C) will not be involved in any heat transfer (D) nothing can be said

23. The radius of a planet is n times the radius of earth, R. A satellite revolves around it in a circle of radius 4nR with angular velocity ω. The acceleration due to gravity on planet's surface is
(A) $R\omega^2$ (B) $16\ R\omega^2$
(C) $32\ nR\omega^2$ (D) $64\ nR\omega^2$.

24. Some successive frequencies of an ideal organ pipe are measured as 150 Hz, 250 Hz, 350 Hz. Velocity of sound is 340 m/s.
(A) it must be an open organ pipe of length 3.4 m
(B) it must be an open organ pipe of length 6.8 m
(C) it must be an closed organ pipe of length 1.7 m
(D) it must be an closed organ pipe of length 3.4 m.

Space for Rough work

25. Two identical blocks are kept on a frictionless horizontal table and connected by a spring of stiffness 'k' and of original length ℓ_0. A total charge Q is distributed on the blocks such that maximum elongation of spring, at equilibrium, is equal to x. Value of Q is

(A) $2\ell_0\sqrt{4\pi\varepsilon_0 k(\ell_0 + x)}$ (B) $2x\sqrt{4\pi\varepsilon_0 k(\ell_0 + x)}$

(C) $2(\ell_0 + x)\sqrt{4\pi\varepsilon_0 kx}$ (D) $(\ell_0 + x)\sqrt{4\pi\varepsilon_0 kx}$.

26. At t = 0, light of intensity 10^{12} photons/s–m^2 of energy 6eV per photon start falling on a plate with work function 2.5 eV. If area of the plate is 2×10^{-4} m^2 and for every 10^5 photons one photoelectron is emitted, charge on the plate at t = 25 s is

(A) 8×10^{-15} C (B) 4×10^{-15} C

(C) 12×10^{-15} C (D) 16×10^{-15} C.

27. Two resistances are measured in Ohm.

$R_1 = 3\Omega \pm 1\%$

$R_2 = 6\Omega \pm 2\%$

When they are connected in parallel, the percentage error in equivalent resistance is

(A) $\frac{14}{3}\%$ (B) 1%

(C) 2% (D) $\frac{4}{3}\%$

28. If λ_{min} is minimum wavelength produced in X–ray tube and $\lambda_{K\alpha}$ is the wavelength of K_α line. As the operating voltage is increased

(A) $\lambda_{K\alpha}$ increases (B) $\lambda_{K\alpha}$ decreases

(C) $(\lambda_{K\alpha} - \lambda_{min})$ increases (D) $(\lambda_{K\alpha} - \lambda_{min})$ decreases.

29. The filament current in the electron gun of a Coolidge tube is increased while the potential difference used to accelerate the electron is decreased. As a result in the emitted radition.

(A) The intensity decreases while the minimum wavelength increases.

(B) The intensity increases while the minimum wavelength decreases.

(C) The intensity as well as the minimum wavelength increases.

(D) The intensity as well as the minimum wavelength decreases.

30. Monochromatic light of wavelength 900 nm is used in a young's double slit experiment. One of the slits is covered by a transparent sheet of thickness 1.8×10^{-5} m made of material of refractive index 1.6. How many fringes will shift due to introduction of the sheet.

(A) 18 (B) 12

(C) 10 (D) 6

Space for Rough work

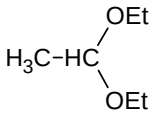
Chemistry

PART – II

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Compound  and $H_3C—CH_2—O—CH_2—CH_3$ can not be differentiated by

(p) (Acetal) (q)

(A) $H_3O^{\oplus}$, Na (B) $H_3O^{\oplus}$, Tollens test

(C) $H_3O^{\oplus}$, Fehling test (D) None of the above

2. If Ksp of AgX are in the order AgCl > AgBr > AgI, then what will be the order of $E^{\circ}_{X^- / AgX / Ag}$ for different half cells having different silver halides

(A) AgCl < AgBr < AgI (B) AgCl > AgBr > AgI

(C) AgCl = AgBr = AgI (D) E^0 is not related to Ksp

3. Which of the following compounds was unable to respond chromyl chloride Test?

(A) NaCl (B) $SnCl_2$

(C) $SnCl_4$ (D) AuCl

4. The activation energy for the decomposition of hydrogen iodide at 581 K is 209.5 cal. Then the percentage of the molecules crossing over the barrier is:
Given that antilog (-1.07828) = 0.0835

(A) 83.5% (B) 8.35%

(C) 40% (D) 60%

5. A mixture of SO_3, SO_2 and O_2 gases is maintained at equilibrium in 10 litre flask at a temperature at which K_c for the reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is 100. If number of moles of SO_2 and SO_3 are same at equilibrium, the number of moles of O_2 present is:

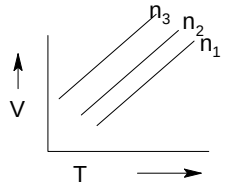
(A) 0.01 (B) 0.1

(C) 1 (D) 10

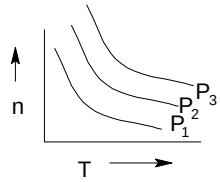
Space for Rough work

6. Two containers A and B (having same volume) containing N_2 at pressure P_1 and the temperature is maintained T_1, if these two are now connected with a narrow tube and one container is still at T_1 and other changes to T_2 then the new pressure in the vessel can be given as
(A) $2P_1/(T_1+T_2)$ (B) $2P_1T_2/(T_1+T_2)$
(C) $P_1T_2/2(T_1+T_2)$ (D) $T_1/2P_1T_2$

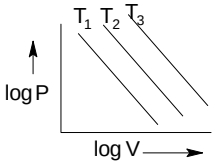
7. Which of the following curves does not represent the behaviour of an ideal gas

(A) 

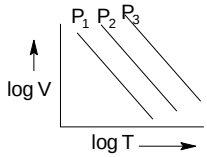
n_1, n_2 & n_3 are no. of moles of the gases

(B) 

at constant V

(C) 

at constant n

(D) 

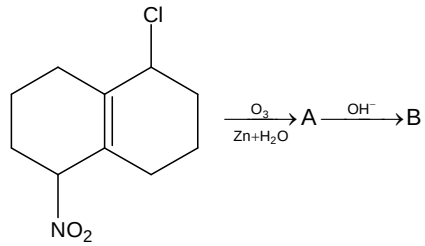
at constant P and constant n

8. A solution of A and B with 30 mole percent of A is in equilibrium with its vapour which contains 60 mole percent of A. Assuming ideality of the solution and the vapour. The ratio of vapour pressure of pure A to that of pure B will be
(A) 7:2 (B) 7:4
(C) 4:7 (D) 1:2

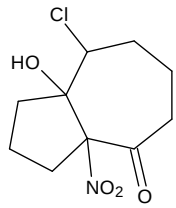
9. Potassium halide has unit cell length of 0.5 $\mathring{A}$ lesser than that for the corresponding Rb salt of the same halogen. What is the ionic radius of K^+ if that of Rb^+ is 1.58 $\mathring{A}$?
(A) 2.08 $\mathring{A}$ (B) 1.33 $\mathring{A}$
(C) 1.08 $\mathring{A}$ (D) 1.28 $\mathring{A}$

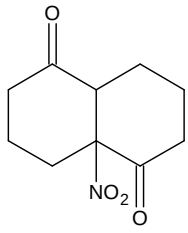
10. In the process of extraction of Al from bauxite coke powder is spreaded over the molten electrolyte to:
(A) Prevent the loss of heat by radiation from the surface
(B) Prevent the corrosion of graphite anode
(C) Prevent oxidation of molten aluminium by air
(D) Both (A) and (C)

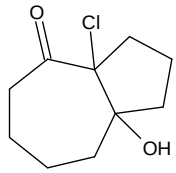
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11. 

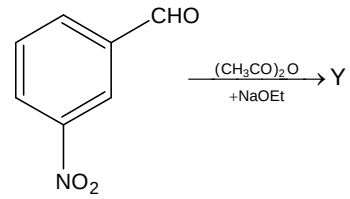
The end product (B) would be

(A) 

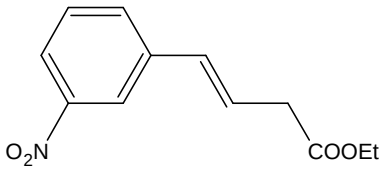
(B) 

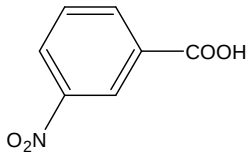
(C) 

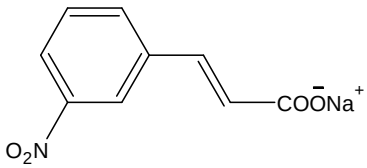
(D) Both A & B

12. 

Y would be

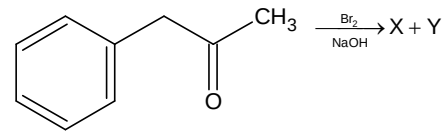
(A) 

(B) 

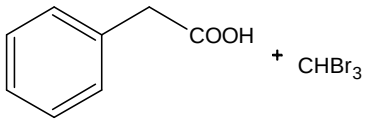
(C) 

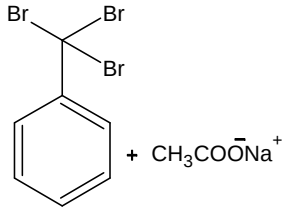
(D) None

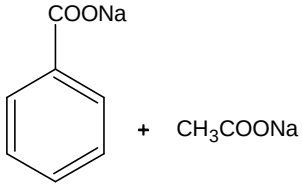
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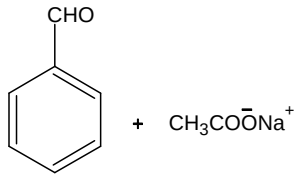
13. 

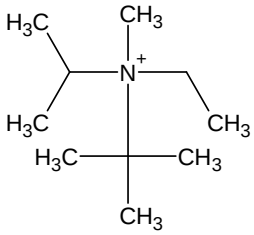
X and Y would be

(A) 

(B) 

(C) 

(D) 

14. 

A(alone), A is

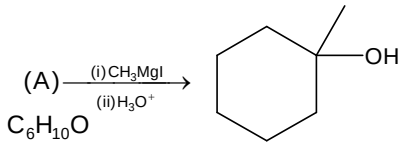
(A) $CH_2 = CH_2$

(B) $CH_3 - CH = CH_2$

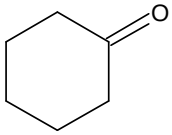
(C) $CH_3 - \overset{CH_3}{\overset{|}{C}} = CH_2$

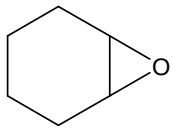
(D) $CH_3 - \overset{CH_3}{\overset{|}{C}} = CH - N(CH_3) - CH_2 - CH_3$

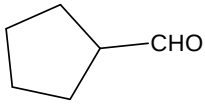
Space for Rough work

15. 

The structure of the compound (A) is

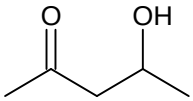
(A) 

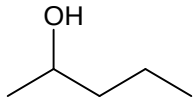
(B) 

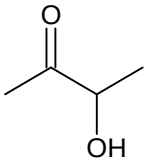
(C) 

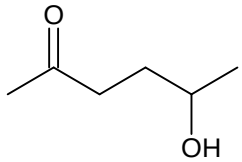
(D) All the above

16. Which one of the following will most readily be dehydrated in acidic condition?

(A) 

(B) 

(C) 

(D) 

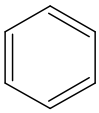
17. At Temperature T, a compound $AB_2(g)$ dissociates according to the reaction.
$2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$ with a degree of dissociation 'x', which is small compared to unity. Then the expression for 'x' is: [k_p is equilibrium constant, P is total pressure].

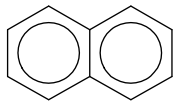
(A) $\left(2k_p/P\right)^3$ (B) $\left(2k_p/P\right)^2$

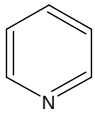
(C) $\left(2k_p/P\right)^{\frac{1}{3}}$ (D) $\left(2k_p/P\right)^{\frac{1}{2}}$

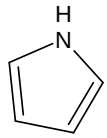
Space for Rough work

18. Which of the following is most reactive towards electrophilic aromatic substitution

(A) 

(B) 

(C) 

(D) 

19. Consider the reaction

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} + \text{Heat}$$

What will happen if 0.5 mole of Helium gas is introduced into the vessel so that the temperature and pressure remain constant?

(A) the equilibrium will shift to right
(B) the equilibrium will shift to left
(C) no change
(D) equilibrium constant will increase

20. The bond angle around central atom is highest in

(A) BBr_3
(B) CS_2
(C) SO_2
(D) SF_4

21. Acetone an heating with conc. H_2SO_4 mainly gives

(A) Mesitylene
(B) Mesityl oxide
(C) Toluene
(D) Xylene

22. Which is not dissolved by dil. HCl?

(A) ZnS
(B) MnS
(C) $BaSO_3$
(D) $BaSO_4$

Space for Rough work

23. When electricity is passed through aqueous solution of aluminium chloride, 13.5g of Al are deposited. The number of faradays of electricity passed must be
(A) 2 (B) 1.5
(C) 1.0 (D) 0.5

24. Which one of the following statement is incorrect?
(A) A solution freezes at lower temperature than the pure solvent.
(B) A solution boils at a higher temperature than the pure solvent.
(C) 0.1 M NaCl solution and 0.1M sugar solution have the same boiling point.
(D) Osmosis cannot take place without a semi permeable membrane.

25. The metallurgical process in which a metal is obtained in a fused state is called
(A) Smelting (B) Roasting
(C) Calcination (D) Froth floatation

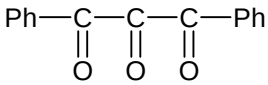
26. Which one of the following doesn't react with NH_3 ?
(A) Benzaldehyde (B) Acetone
(C) Formaldehyde (D) Glucose

27. The pair of compound which cannot exist together in solution is
(A) $NaHCO_3$ and NaOH (B) Na_2CO_3 and $NaHCO_3$
(C) Na_2CO_3 and NaOH (D) $NaHCO_3$ and NaCl

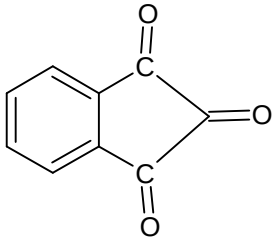
28. $X^+ Y^-$ crystallises in a bcc lattice in which the radius of $X^+ = 0.4\,\mathring{A}$. If all the X^+ ions are removed and another cation Z^+ is inserted, what should be the maximum radius of the cation Z^+ to crystallise in a fcc lattice with Y^- :
(A) $0.546\,\mathring{A}$ (B) $0.010\,\mathring{A}$
(C) $0.400\,\mathring{A}$ (D) $0.226\,\mathring{A}$

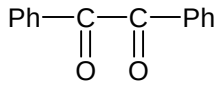
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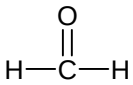
29. Which will form(s) stable gem-diol:

(I) 

(II) CCl_3CHO

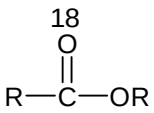
(III) 

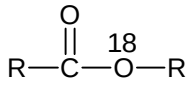
(IV) 

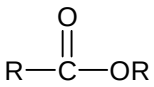
(V) 

(A) I, II, III, V only
(B) I, II, III, IV only
(C) II, III, IV only
(D) II, III, IV, V only

30. The reaction of $R-\overset{\overset{18}{O}}{\overset{\|}{C}}-OH$ with H^+/ROH will give:

(I) 

(II) 

(III) 

(A) I, II only
(B) I, III only
(C) I, II and III only
(D) I only

Space for Rough work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. If the graph of the function $f(x) = \frac{a^x - 1}{x^n(a^x + 1)}$ is symmetrical about y–axis, then n equals

(A) 2 (B) $\frac{2}{3}$

(C) $\frac{1}{4}$ (D) $-\frac{1}{3}$

2. $\lim_{x\to\infty} \frac{\cot^{-1}\left(x^{-a}\log_a x\right)}{\sec^{-1}\left(a^x \log_a a\right)}$ (a > 1) is equal to

(A) 2 (B) 1
(C) $\log_a 2$ (D) 0

3. If $f(x) = \begin{cases} \sin x & , \ x < 0 \\ \cos x - |x-1| & , \ x \geq 0 \end{cases}$ then g(x) = f(|x|) is non–differentiable for

(A) no value of x (B) exactly one value of x
(C) exactly 3 values of x (D) none of these

4. The radius of a right circular cylinder increases at the rate of 0.1 cm/min, and the height decreases at the rate of 0.2 cm/min. The rate of change of the volume of the cylinder, in cm^3/min, when the radius is 2 cm and the height is 3 cm is

(A) -2π (B) $-\frac{8\pi}{5}$

(C) $-\frac{3\pi}{5}$ (D) $\frac{2\pi}{5}$

Space for rough work

5. If $3(a + 2c) = 4(b + 3d)$, then the equation $ax^3 + bx^2 + cx + d = 0$ will have
(A) no real solution (B) at least one real root in (–1, 0)
(C) at least one real root in (0, 1) (D) none of these

6. If $f : R \to R$ and $g : R \to R$ are two functions such that $f(x) + f''(x) = -x\, g(x)f'(x)$ and $g(x) > 0\ \forall\ x \in R$, then the functions $f^2(x) + (f'(x))^2$ has
(A) a maxima at x = 0 (B) a minima at x = 0
(C) a point of inflection at x = 0 (D) none of these

7. The fuel charges for running a train are proportional to the square of the speed generated in km per hour, and the cost is Rs. 48 at 16 km per hour. If the fixed charges amount to Rs. 300 per hours, the most economical speed is
(A) 60 kmph (B) 40 kmph
(C) 48 kmph (D) 36 kmph

8. If the function $f(x) = \dfrac{t + 3x - x^2}{x - 4}$, where t is a parameter that has a minimum and maximum, then the range of values of t is
(A) (0, 4) (B) $(0, \infty)$
(C) $(-\infty, 4)$ (D) $(4, \infty)$

9. If $4\hat{i} + 7\hat{j} + 8\hat{k}$, $2\hat{i} + 3\hat{j} + 4\hat{k}$ and $2\hat{i} + 5\hat{j} + 7\hat{k}$ are the position vectors of the vertices A, B and C, respectively, of triangle ABC, the position vector of the point where the bisector of angle A meets BC, is
(A) $\dfrac{2}{3}\left(-6\hat{i} - 8\hat{j} - 6\hat{k}\right)$ (B) $\dfrac{2}{3}\left(6\hat{i} + 8\hat{j} + 6\hat{k}\right)$
(C) $\dfrac{1}{3}\left(6\hat{i} + 13\hat{j} + 18\hat{k}\right)$ (D) $\dfrac{1}{3}\left(5\hat{j} + 12\hat{k}\right)$

10. $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$. A vector coplanar with $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is magnitude $\sqrt{\dfrac{2}{3}}$ is
(A) $2\hat{i} + 3\hat{j} - 3\hat{k}$ (B) $-2\hat{i} - \hat{j} + 5\hat{k}$
(C) $2\hat{i} + 3\hat{j} + 3\hat{k}$ (D) $2\hat{i} + \hat{j} + 5\hat{k}$

Space for rough work

11. If $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 0$, then $\left(\vec{a} \times \left(\vec{a} \times \left(\vec{a} \times \vec{b}\right)\right)\right)$ is equal to

(A) $48\hat{b}$ (B) $-48\hat{b}$

(C) $48\hat{a}$ (D) $-48\hat{a}$

12. If $\vec{r} = x_1(\vec{a} \times \vec{b}) + x_2(\vec{b} \times \vec{a}) + x_3(\vec{c} \times \vec{d})$ and $4[\vec{a}\ \vec{b}\ \vec{c}] = 1$, then $x_1 + x_2 + x_3$ is equal to

(A) $\frac{1}{2}\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$ (B) $\frac{1}{4}\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$

(C) $2\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$ (D) $4\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$

13. The point of intersection of the line passing through (0, 0, 1) and intersecting the lines $x + 2y + z = 1$, $-x + y - 2z = 2$ and $x + y = 2$, $x + z = 2$ with xy plane is

(A) $\left(\frac{5}{3}, -\frac{1}{3}, 0\right)$ (B) (1, 1, 0)

(C) $\left(\frac{2}{3}, -\frac{1}{3}, 0\right)$ (D) $\left(-\frac{5}{3}, \frac{1}{3}, 0\right)$

14. The lines which intersect the skew lines $y = mx$, $z = c$; $y = -mx$, $z = -c$ at the x–axis lies on the surface

(A) $cz = mxy$ (B) $xy = cmz$

(C) $cy = mxz$ (D) none of these

15. A light ray gets reflected from the line $x = -2$. If the reflected ray touches the circle $x^2 + y^2 = 4$ and point of incidence is $(-2, -4)$, the equation of incident ray is

(A) $4y + 3x + 22 = 0$ (B) $3y + 4x + 20 = 0$

(C) $4y + 2x + 20 = 0$ (D) $y + x + 6 = 0$

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16. If S_1 and S_2 are the foci of the hyperbola whose transverse axis length is 4 and conjugate axis length is 6, S_3 and S_4 are the foci of the conjugate hyperbola, then the area of the quadrilateral $S_1S_2S_3S_4$ is
(A) 24 (B) 26
(C) 22 (D) none of these

17. $\sum_{r=1}^{n} \sin^{-1}\left(\frac{\sqrt{r}-\sqrt{r-1}}{\sqrt{r(r+1)}}\right)$ is equal to
(A) $\tan^{-1}\left(\sqrt{n}\right)-\frac{\pi}{4}$ (B) $\tan^{-1}\left(\sqrt{n+1}\right)-\frac{\pi}{4}$
(C) $\tan^{-1}\left(\sqrt{n}\right)$ (D) $\tan^{-1}\left(\sqrt{n+1}\right)$

18. In triangle ABC, a = 5, b = 4 and c = 3. G is the centroid of the triangle. Circumradius of triangle GAB is equal to
(A) $2\sqrt{13}$ (B) $\frac{5}{12}\sqrt{13}$
(C) $\frac{5}{3}\sqrt{13}$ (D) $\frac{3}{2}\sqrt{13}$

19. 1, z_1, z_2, z_3, ….., z_{n-1} are the nth roots of unity, then the value of $\frac{1}{3-z_1}+\frac{1}{3-z_2}+.....+\frac{1}{3-z_{n-1}}$ is equal to
(A) $\frac{n\cdot 3^{n-1}}{3^n-1}+\frac{1}{2}$ (B) $\frac{n\cdot 3^{n-1}}{3^n-1}-1$
(C) $\frac{n\cdot 3^{n-1}}{3^n-1}+1$ (D) none of these

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20. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{r}{1\times3\times5\times7\times.....\times(2r+1)}$ is equal to

(A) $\frac{1}{3}$ (B) $\frac{3}{2}$

(C) $\frac{1}{2}$ (D) none of these

21. If n integers taken at random are multiplied together then the probability that the last digit of the product is 1, 3, 7 or 9 is

(A) $\frac{2^n}{5^n}$ (B) $\frac{4^n-2^n}{5^n}$

(C) $\frac{4^n}{5^n}$ (D) none of these

22. The number of distinct real roots of $\begin{vmatrix}\sin x & \cos x & \cos x\\ \cos x & \sin x & \cos x\\ \cos x & \cos x & \sin x\end{vmatrix}=0$ in the interval $-\frac{\pi}{4}\le x\le\frac{\pi}{4}$ is

(A) 0 (B) 2

(C) 1 (D) 3

23. Area of the region bounded by the curves $y = 2^x$, $y = 2x - x^2$, $x = 0$ and $x = 2$ is given by

(A) $\frac{3}{\log 2}-\frac{4}{3}$ (B) $\frac{3}{\log 2}+\frac{4}{3}$

(C) $3\log 2-\frac{4}{3}$ (D) none of these

24. $\sim[(\sim p)\wedge q]$ is equal to

(A) $p\vee(\sim q)$ (B) $p\vee q$

(C) $p\wedge(\sim q)$ (D) $\sim p\wedge\sim q$

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25. The angle of elevation of the top of the tower observed from each of the three points A, B, C on the ground, forming a triangle is the same angle α. If R is the circum–radius of the ΔABC, then the height of the tower is
(A) R sin α (B) R cos α
(C) R cot α (D) R tan α

26. The mean of two samples of size 200 and 300 were found to be 25, 10 respectively. Their standard deviations were 3 and 4 respectively. The variance of combined sample of size 500 is
(A) 64 (B) 65.2
(C) 67.2 (D) 64.2

27. If $I = \int_0^{\pi/2} \frac{dx}{\sqrt{1+\sin^3 x}}$, then
(A) $I \geq \frac{\pi}{2}$ (B) $I \geq \frac{\pi}{2\sqrt{2}}$
(C) $I \leq \frac{\pi}{2\sqrt{2}}$ (D) none of these

28. The solution of $x = 1+\left(xy\frac{dy}{dx}\right)+\frac{x^2y^2}{2!}\left(\frac{dy}{dx}\right)^2+\frac{(xy)^3}{3!}\left(\frac{dy}{dx}\right)^3+.....$ is
(A) y = ln x + c (B) $y = (\ln x)^2 + c$
(C) $y = \pm\sqrt{(\ln x)^2 + c}$ (D) none of these

29. A parabola $y = ax^2 + bx + c$ crosses the x–axis at (α, 0) and (β, 0) both to the right of the origin. A circle also passes through these two points. The length of a tangent from the origin to the circle is
(A) $\sqrt{\frac{bc}{a}}$ (B) ac^2
(C) $\frac{b}{a}$ (D) $\sqrt{\frac{c}{a}}$

30. If $f(x) = 1 – x + x^2 – x^3 + ... –x^{15} + x^{16} – x^{17}$, then the coefficient of x^2 in f(x – 1) is
(A) 826 (B) 816
(C) 822 (D) none of these

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